

MURCIA, Spain

WMO: 084300

Lat: 38.0019N

Lon: 1.1708W

Elev: 62

StdP: 100.58

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.7	4.1	-7.2	2.1	10.2	-4.7	2.6	10.4	9.4	15.0	8.2	14.7	1.3	270	0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.1	36.6	22.1	35.1	21.8	33.8	21.7	24.9	31.8	24.1	30.6	23.5	29.7	3.8	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	18.0	27.1	22.4	17.2	26.8	21.7	16.5	26.6	75.7	31.6	72.8	30.7	70.3	29.7	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.7	6.6	5.8	-0.4	40.0	1.4	2.4	-1.4	41.7	-2.2	43.1	-3.0	44.5	-4.0	46.3
			-1.9	26.4	1.6	1.3	-3.1	27.3	-4.0	28.1	-5.0	28.9	-6.1	29.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.2	11.6	12.6	14.8	17.2	20.8	25.1	27.9	28.0	24.4	20.6	15.1	12.1
	DBStd	6.39	2.71	2.75	2.82	2.56	2.56	2.43	1.83	1.80	2.38	2.58	3.00	2.60
	HDD10.0	35	13	8	1	0	0	0	0	0	0	0	2	11
	HDD18.3	852	209	162	115	50	7	0	0	0	0	9	104	194
	CDD10.0	3399	63	80	149	217	336	454	555	557	432	328	153	74
	CDD18.3	1174	0	1	5	17	85	204	297	298	182	79	6	0
	CDH23.3	11607	2	7	74	169	693	2010	3349	3312	1504	450	35	1
CDH26.7	5192	0	1	13	25	193	906	1739	1647	562	103	3	0	
Wind	WSAvg	2.3	2.1	2.3	2.5	2.5	2.6	2.7	2.7	2.5	2.1	1.8	2.0	1.8
Precipitation	PrecAvg	283	16	26	30	32	32	23	2	7	23	41	32	23
	PrecMax	575	59	107	146	98	130	90	9	31	131	175	132	75
	PrecMin	113	1	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	114	16	27	31	26	26	24	3	10	31	47	32	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.8	24.4	28.3	29.2	33.1	37.4	39.2	38.7	35.3	31.4	26.8	22.6
		MCWB	13.8	13.7	15.4	16.9	19.0	21.0	22.8	23.1	21.2	19.5	16.6	14.5
	2%	DB	20.5	22.0	25.3	26.9	30.5	34.9	36.7	36.5	33.1	29.2	24.1	20.3
		MCWB	12.7	13.1	14.7	15.9	18.1	20.6	22.1	22.6	21.3	19.3	15.7	13.6
	5%	DB	18.7	20.2	23.2	25.2	28.7	33.1	35.1	34.9	31.4	27.6	22.2	18.7
		MCWB	11.6	12.4	13.9	15.1	17.5	20.3	21.9	22.3	21.0	19.0	14.8	12.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	16.8	18.3	21.3	23.5	27.1	31.4	33.7	33.5	29.9	25.9	20.4	17.0
		MCWB	10.8	11.5	13.0	14.5	17.0	19.8	21.8	22.3	20.7	18.3	14.2	11.9
	0.4%	WB	14.7	15.6	17.0	18.4	21.1	24.3	25.9	25.9	24.8	21.8	18.3	16.1
		MCDB	21.5	21.5	25.4	26.4	29.5	32.4	34.2	34.1	30.4	27.1	23.1	20.2
	2%	WB	13.4	14.0	15.5	17.1	19.6	22.6	24.7	25.0	23.5	20.8	17.1	14.6
		MCDB	19.0	19.9	23.2	24.7	27.2	30.3	32.1	31.8	29.0	26.0	21.5	18.4
Mean Daily Temperature Range	5% DB	WB	12.4	13.0	14.5	16.1	18.7	21.7	24.0	24.3	22.7	20.2	16.2	13.5
		MCDB	17.0	18.6	21.3	22.8	25.6	29.1	31.0	30.5	28.1	25.1	20.2	17.4
		WB	11.5	12.1	13.6	15.2	17.9	20.9	23.3	23.8	22.0	19.5	15.3	12.6
	5% WB	MCDB	15.7	17.0	19.5	21.4	24.6	28.3	29.9	29.6	27.1	24.0	19.2	16.2
Clear-Sky Solar Irradiance	taub	0.314	0.341	0.391	0.404	0.432	0.455	0.461	0.465	0.433	0.396	0.347	0.322	
	taud	2.462	2.403	2.258	2.253	2.201	2.159	2.153	2.146	2.239	2.335	2.438	2.462	
	Ebn at Noon	868	884	866	875	853	831	822	809	813	810	819	830	
	Edh at Noon	82	99	126	134	144	150	150	148	127	105	84	77	
All-Sky Solar Radiation	RadAvg	2.56	3.38	4.58	5.66	6.68	7.54	7.48	6.51	5.07	3.73	2.67	2.22	
	RadStd	0.25	0.35	0.38	0.41	0.47	0.24	0.17	0.23	0.27	0.29	0.25	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.36	N/A	N/A	+0.93	+0.79	+0.58	N/A	N/A	+185	+132	(46)	
(47)	Regional (5 neighbors)	N/A	N/A	-1.30	+0.51	N/A	N/A	N/A	N/A	N/A	+58	(47)	

Nomenclature: See separate page